PRANJAL SINGH SESODIA

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EDUCATION

Netaji Subhas University of Technology (NSUT) **Aug 2024 - Aug 2028 (Expected)**

B.Tech. Mechanical Engineering | New Delhi, India CGPA: 8.07/10 (Dept. Rank: 19/215, **Top 9%**)

- **Coursework:** Control Systems, Theory of Machines, Strength of Materials, Fluid Mechanics & Machines; ongoing: Robotics, Machine Design.

RESEARCH EXPERIENCE

R&D Intern, mPRAGATI, Indian Institute of Technology Delhi **Jun 2026 - Present**

New Delhi, India | Supervisor: Dr. Dinesh Kalyanasundaram

- Derived complete **6-DOF DH kinematics, geometric Jacobian, singularity locus** and **damped least-squares (DLS) inverse kinematics** of robotic arm in MATLAB, validated against the Robotics System Toolbox.
- Implemented **30 Hz image-based visual servoing (IBVS)** in Python (OpenCV, MediaPipe) on a custom **encoder-free 4-DOF manipulator** that keeps a surgical camera locked on its target under partial occlusion; integrated eye-to-hand joint-angle estimation (developed at QuEST Lab) with EMA filtering.
- Designed coarse-fine tilt allocation across wrist, elbow and shoulder with automatic wrist recentering, **eliminating jerk during operations**.
- Designed a **5:1 cycloidal reducer** in SolidWorks for the 6-DOF stage with epitrochoid profile (interference-free over 360° input); prototyped in FDM, SLA and resin with **process-specific clearances of 0.15-0.3 mm**.
- Diagnosed a **trajectory mismatch in a patented total elbow implant** with ANSYS Multibody Dynamics: the specified elliptical ulnar path (2-4 mm offset) simulated as two circular arcs with an offset jump at 90° flexion; **redesigned the hinge kinematics**.
- Re-synthesized the links of a polycentric four-bar prosthetic knee to satisfy the **Grashof condition**, **restoring full flexion**.

Research Intern, QuEST Lab, IIT Delhi **Dec 2025 - May 2026**

New Delhi, India | Supervisor: Dr. Jay Dhariwal

- Developed a **contactless joint-angle measurement algorithm** (Python, OpenCV, AlphaPose) that tracks surgeons through occlusion via limb-line extrapolation; now adapted for the mPRAGATI arm.
- Developed and deployed a **7 cm-baseline stereo rig** (OpenCV stereo calibration and triangulation) for markerless surgical motion capture; triangulated depth reached **95-96% accuracy** against reference distances measured with standard instruments.
- Co-developed a **pump and lining system** for saline pressure and fluid management in arthroscopic surgery (four-member team).

PUBLICATIONS

- **Sesodia, P. S.**, & Dabass, V. (2026). Robust Real-Time Levitation Gap Estimation for Hyperloop Pods. *Next Research (Elsevier).* DOI: 10.1016/j.nexres.2026.102384 Real-time air-gap estimation for closed-loop control of open-loop-unstable EMS maglev actuators.

PROJECTS

Tabletop Subsonic Wind Tunnel *NSUT, New Delhi | Supervisor: Dr. Vivek Kumar* **Mar 2026 - Present**

- Designed a **150 mm test section** with a **6:1 Bell-Mehta contraction** refined with **ANSYS Fluent CFD**; cut printed mass with static structural FEA; integrated the motor drive and pressure DAQ.

Continuum Robot | mPRAGATI, IIT Delhi **Jun 2026**

- Designed a **tendon-driven continuum bronchoscope handle** (SolidWorks) with a motorized differential bevel-gear train for **dual-plane tip articulation**.

Team Daedalus Racing *BAJA SAEINDIA | NSUT, New Delhi* **Oct 2025 - Mar 2026**

- Changed the final-drive reduction from **11:1 to 8:1** for low-gradient endurance (**+37.5% top speed** at equal engine RPM); developed a lower-cost catalytic converter, assessing emissions in Ansys Granta Selector (sustainability report, **1st nationally**).

EYEAR: Real-Time Speech-to-Sign Assistive Device | Army Public School, Shankar Vihar **2021 - 2022**

- Built a **real-time speech-to-sign device** on Raspberry Pi (Python); **Indian provisional patent (2022)**; selected by IBM to present it to the then **Union Minister of Education**.

HONORS & LEADERSHIP

- **Awards & Grants: 1st Place Nationally (INR 30,000)**, mBAJA 2026; **NSUT CPVS Research Grant** (INR 10,000); Smart India Hackathon Jr. Finalist (2022).
- Taught a **2-day 3D-printing workshop to 60 students** (NSUT IIF, 2026); **mentored 8 students** to a Li-ion battery recycling prototype (Change.Makers 2.0).

TECHNICAL SKILLS

- **Robotics & Control:** forward and inverse kinematics (DH convention, damped least squares), Jacobian and singularity analysis, image-based visual servoing (IBVS), PID control, Euler-Lagrange dynamics, computed-torque control
- **Mechanical Design & Analysis:** CAD, finite element analysis (FEA), multibody dynamics, computational fluid dynamics (CFD), linkage synthesis, gear design, GD&T, design for manufacturing (DFM), FDM and SLA 3D printing
- **Software:** SolidWorks, MATLAB (Robotics System Toolbox), ANSYS (Mechanical, Multibody Dynamics, Fluent, Granta), Python (NumPy, OpenCV), C
- **Computer Vision:** stereo calibration and triangulation, human pose estimation (MediaPipe, AlphaPose), marker-based pose estimation, monocular depth estimation (Depth Anything V2)